

Aydin Fakhri Kaleybar

Computer Science Researcher | Process Mining | AI & Data-Driven Systems | Software Engineering

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[Google Scholar](#) · [LinkedIn](#) · [ResearchGate](#) · [GitHub](#)

Summary

Computer Science researcher working at the intersection of Process Mining, AI, and sustainable computing, with a focus on healthcare applications. First author of a published systematic review on healthcare process mining, a submitted review proposing a polyglot hybrid database architecture to improve analytical performance while reducing energy and CO₂ footprint, and an in-progress review on Agentic AI in software engineering. Also a contributing author on peer-reviewed research in software defect prediction and metaheuristic optimization. Proficient in Python, SQL, and a range of database and machine-learning tools. Holds M.Sc. and B.Sc. degrees in Software/Computer Engineering from Islamic Azad University, Tabriz Branch.

Educational Background

M.Sc., Computer Engineering, Software Engineering

2023-2025

Department of Computer Engineering, Islamic Azad University, Tabriz Branch, Tabriz, Iran

- Thesis:** Toward Sustainable Healthcare Process Mining: A Polyglot Hybrid Database Architecture for Efficient Analytics and CO₂ Reduction
- Supervised by:** Dr. Mohammadbagher Karimi
- Total GPA:** ?/4 (CGPA: 19.25/20 on the Iranian Scale)
- Ranked among the top 3 students of the cohort by academic performance

B.Sc., Computer Engineering, Information Technology

2018-2022

Department of Computer Engineering, Islamic Azad University, Tabriz Branch, Tabriz, Iran

- Final Project:** ParentConnect; designed and developed a Django-based social platform that enables parents to collaborate, share insights, and exchange feedback on their children's educational and developmental experiences
- Advised by:** Dr. Mohammadbagher Karimi
- CGPA of the Last Two Years:** ?/4 (16.38/20 on the Iranian Scale)

Research Interests

- Healthcare Process Mining
- AI/ML-Driven Software Defect Prediction
- Sustainable & Polyglot Database Architecture Design
- Data-Intensive Software System Design
- Agentic AI & Intelligent Systems in Software Engineering
- Metaheuristic Optimization Algorithms

Academic Experience

Research Assistant | Islamic Azad University, Tabriz

2023- 2025

- Conducted a PRISMA 2020–based systematic review of Process Mining in healthcare, analyzing algorithmic trends, tool use, and clinical applications such as pathway analysis, conformance checking, and bottleneck detection (published in the International Journal of Data Science and Analytics, Springer; first author)
- Designed and evaluated a three-tier polyglot persistence architecture (MySQL, MongoDB, Redis) for healthcare process mining as the first author of the M.Sc. thesis, reducing execution time by ~18.6% and energy consumption by ~14.2% while preserving predictive model quality (F_1 -score = 0.82 ± 0.02), validated on the BPI Challenge 2011 dataset

Independent Research Collaboration | With Faculty Advisors, Islamic Azad University, Tabriz

2025-Present

- Expanded the M.Sc. thesis into a full manuscript, "Toward Sustainable Healthcare Process Mining: A Polyglot Hybrid Database Architecture for Efficient Analytics and CO₂ Reduction," submitted to the International Journal of Sustainable Computing: Informatics and Systems (Elsevier) and currently under peer review (first author)
- Co-authored a systematic review of advanced learning techniques for software defect prediction, covering ensemble learning, deep learning, GNNs, and Transformer-based models, based on 32 primary studies. Currently under peer review at the International Journal of Computational Intelligence Systems, Springer (third author)
- Contributing to a systematic review of Agentic AI and intelligent agents across the software development lifecycle, examining governance, human–agent collaboration, and educational implications for AI-native software engineering (*in preparation; first author*)
- Contributing to a systematic review of the Parrot Optimizer algorithm, covering its evolution, hybridization strategies, and application domains (*in preparation; co-author*)

Publications

[1] **Systematic review: a literature review of process mining in healthcare.**

(2026)

International Journal of Data Science and Analytics

Aydin Fakhri Kaleybar, Mohammadbagher Karimi, Niloofar Naghshi, Bahman Arasteh

[2] **Using Advanced Learning Techniques to Predict Software Defects: A Systematic Literature Review.** (Under Peer Review) (2026)

International Journal of Computational Intelligence Systems

Niloofer Naghshi, Mohammadbagher Karimi, Aydin Fakhri Kaleybar, Bahman Arasteh, Huseyin Kusetogullar

[3] **Toward Sustainable Healthcare Process Mining: A Polyglot Hybrid Database Architecture for Efficient Analytics and CO₂ Reduction.** (Under Peer Review) (2026)

International Journal of Sustainable Computing: Informatics and Systems

Aydin Fakhri Kaleybar, Mohammadbagher Karimi, Bahman Arasteh, Niloofer Naghshi, Huseyin Kusetogullar

[4] **Agentic AI and Methodological Transformation of Software Engineering: A Systematic Review of Intelligent Agents in the Development Lifecycle.** (In Preparation) (2026)

Aydin Fakhri Kaleybar, Mohammadbagher Karimi, Niloofer Naghshi, Bahman Arasteh, Huseyin Kusetogullar

[5] **Parrot Optimizer: A Systematic Review of Algorithmic Evolution, Hybridization Strategies, Application Domains, and Future Directions.** (In Preparation) (2026)

Niloofer Naghshi, Bahman Arasteh, Mohammadbagher Karimi, **Aydin Fakhri Kaleybar**, Huseyin Kusetogullar

Skills

Programming & Backend Development: Python, Django, RESTful API Development, Backend Architecture

Database Systems: MySQL, PostgreSQL (Relational); MongoDB, Redis (NoSQL); TimescaleDB (Time-series); ChromaDB (Vector DB); Polyglot Persistence Design

Process Mining Tools: ProM, Disco, Alpha Miner, Heuristic Miner, Inductive Miner

Data Analysis & Machine Learning: pandas, scikit-learn, matplotlib, Feature Engineering, Predictive Modeling

Research & Scientific Computing: Systematic Literature Review (PRISMA), Reproducible Experiments, Statistical Analysis

Development & Infrastructure Tools: Git, Docker, Linux

Languages

- **English:** Full Professional Proficiency (scientific writing and academic communication)
- **Persian:** Native
- **Azerbaijani:** Native
- **Turkish (Istanbul Turkish):** Intermediate-Advanced (B2)

References

- **Mohammadbagher Karimi**, Member of Faculty, Computer Engineering Department, Islamic Azad University, Tabriz Branch (IAUT) — *contact details available upon request*
- **Bahman Arasteh**, Associate Professor, Department of Software Engineering, Istinye University, Istanbul, Turkey — *contact details available upon request*